

Qwen / Qwen-AgentWorld-35B-A3B

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image-text-to-text qwen world-model agent environment-simulation conversational
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Qwen-AgentWorld-35B-A3B



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This repository contains the model weights and configuration files for **Qwen-AgentWorld-35B-A3B**, a native language world model trained for agentic environment simulation.

These artifacts are compatible with Hugging Face Transformers, vLLM, SGLang, etc.

Qwen-AgentWorld is the first language world model to cover seven agent interaction domains within a single model. It simulates agentic environments via long chain-of-thought reasoning, predicting the next environment state given an agent's action and interaction history. Trained through a three-stage pipeline — CPT injects

Downloads last month

34,371



Safetensors

Model size 35B params

Tensor type BF16 Chat template

Files info

Inference Providers

Text Generation

This model isn't deployed by any Inference Provider.

10 Ask for provider support

Model tree for Qwen/Qwen-AgentWo...

Base model [Qwen/Qwen3.5-35B-A3B-Ba...](#)

Finetuned (54) [this model](#)

Adapters 1 model

Finetunes 7 models

Merges 1 model

Quantizations 50 models

environment knowledge, SFT activates next-state-prediction reasoning, RL sharpens simulation fidelity — Qwen-AgentWorld is a **native world model**: environment modeling is the training objective from the CPT stage onward, not a post-hoc add-on.

Highlights

- **Seven Unified Domains.** A single model covers MCP (tool calling), Search, Terminal, SWE (software engineering), Android, Web, and OS — spanning both text and GUI interaction environments.
- **Native World Model.** Environment modeling from CPT onward, not post-hoc adaptation on a general-purpose LLM.
- **Generalizable, Scalable & Controllable Simulator.** Zero-shot generalization to OOD environments (e.g., OpenClaw); controllable perturbations and fictional-world construction surpass real-environment training.
- **Agent Foundation Model.** LWM RL warm-up on single-turn, non-agentic trajectories transfers to multi-turn, tool-calling agentic tasks across 7 benchmarks, including 3 entirely out-of-domain.

Model Overview

- Type: Causal Language Model (Language World Model)
- Base Model: Qwen3.5-35B-A3B-Base

 **Dataset used to train Qwen/Qwen-A...**

 Qwen/AgentWorldBench

 Viewer •  2.17k •  1.47k •  60

 **Spaces using Qwen/Qwen-AgentW...** 5

 ginigen-ai/Metacognition-Leaderboa...

 geraldamasi/agentworld-demo

 chabab/qwen-agentworld-35b-zero...

 tuandunghcmut/qwen-agentworld-a...

 charbel-malo/qwen-agentworld-35b...

 **Collection including Qwen/Qwen-Ag...**

Qwen-AgentWorld  Collection

3 items • Updated 7 days ago •  64

 **Paper for Qwen/Qwen-AgentWorld-...**

Qwen-AgentWorld: Language Worl...

 Paper • 2606.24597 • Pub... •  144

- Training Stage: Continual Pre-Training (CPT) → Supervised Fine-Tuning (SFT) → Reinforcement Learning (RL, GSPO)
- Number of Parameters: 35B in total and 3B activated
- Hidden Dimension: 2048
- Token Embedding: 248320 (Padded)
- Number of Layers: 40
- Hidden Layout: $10 \times (3 \times (\text{Gated DeltaNet} \rightarrow \text{MoE}) \rightarrow 1 \times (\text{Gated Attention} \rightarrow \text{MoE}))$
- Gated DeltaNet:
 - Number of Linear Attention Heads: 32 for V and 16 for QK
 - Head Dimension: 128
- Gated Attention:
 - Number of Attention Heads: 16 for Q and 2 for KV
 - Head Dimension: 256
 - Rotary Position Embedding Dimension: 64
- Mixture Of Experts
 - Number of Experts: 256
 - Number of Activated Experts: 8 Routed + 1 Shared
 - Expert Intermediate Dimension: 512
- Context Length: 262,144 tokens
- Disclaimer: No outputs from external API services are included in the training pipeline.

Performance

AgentWorldBench (Open-Ended Evaluation)

Five-dimensional rubric mean per domain,
normalized to 0-100 scale.

Model	MCP	Search	Term.	SWE	Android
GPT-5.4	70.10	37.26	53.69	66.29	60.00
Claude Opus 4.8	54.93	35.14	59.18	64.10	61.50
Claude Opus 4.6	69.90	29.30	57.51	64.55	61.74
Gemini 3.1 Pro	59.07	30.21	52.47	59.07	61.40
Claude Sonnet 4.6	70.00	28.79	56.98	64.52	58.03
DeepSeek-V4-Pro	63.27	27.61	51.26	59.44	55.17
GLM-5.1	67.60	22.46	47.32	52.07	59.10
Kimi K2.6	65.23	27.48	52.54	58.77	58.93
MiniMax-M2.7	55.82	27.30	41.62	37.44	52.40
Qwen3.5-35B-A3B	57.87	25.98	46.13	47.58	53.18
Qwen3.5-397B-A17B	68.31	30.81	55.30	64.44	54.90
Qwen3.6-Plus	55.28	21.94	50.58	59.08	57.65
Qwen-AgentWorld-35B-A3B	64.79	36.69	53.96	65.63	58.17

Model	MCP	Search	Term.	SWE	Androi
Qwen-AgentWorld-397B-A17B	68.24	37.82	57.73	68.49	60.20

Quickstart

Deployment

Qwen-AgentWorld-35B-A3B can be served via APIs with popular inference frameworks. In the following, we show example commands to launch OpenAI-compatible API servers.

The model has a default context length of 262,144 tokens. If you encounter out-of-memory (OOM) errors, consider reducing the context window. However, because Qwen-AgentWorld leverages extended context for multi-turn environment simulation, we advise maintaining a context length of at least 128K tokens.

SGLang

SGLang is a fast serving framework for large language models.

```
python -m sglang.launch_server \
  --model-path Qwen/Qwen-AgentWorld-35B-A3B \
  --port 8000 \
  --tp-size 4 \
  --context-length 262144 \
  --reasoning-parser qwen3
```

An OpenAI-compatible API will be available at `http://localhost:8000/v1`.

vLLM

vLLM is a high-throughput and memory-efficient inference engine for LLMs.

```
vllm serve Qwen/Qwen-AgentWorld-35B-A3B \  
  --port 8000 \  
  --tensor-parallel-size 4 \  
  --max-model-len 262144 \  
  --reasoning-parser qwen3 \  
  --language-model-only \  
  --trust-remote-code
```

“The `--language-model-only` flag is required because the model architecture includes visual component definitions but the checkpoint only contains language model weights. Without this flag, vLLM will attempt to initialize visual modules and fail.”

An OpenAI-compatible API will be available at <http://localhost:8000/v1>.

Inference with Transformers

```
from transformers import AutoModelForCausalLM  
  
model_name = "Qwen/Qwen-AgentWorld-35B-A3B"  
tokenizer = AutoTokenizer.from_pretrained(model_name)  
model = AutoModelForCausalLM.from_pretrained(  
    model_name,  
    torch_dtype="auto",  
    device_map="auto",  
)  
  
messages = [  
    {  
        "role": "system",
```

```

        "content": "You are a language world
                    "Given the user's command

    },
    {
        "role": "user",
        "content": "Action: execute_bash\nCor
    }
]

```

```

text = tokenizer.apply_chat_template(messages,
inputs = tokenizer([text], return_tensors="pt")
outputs = model.generate(**inputs, max_new_tokens=100)
response = tokenizer.decode(outputs[0][inputs.shape[1]:])
print(response)

```

Using via the Chat Completions API

```

from openai import OpenAI

client = OpenAI(
    base_url="http://localhost:8000/v1",
    api_key="EMPTY",
)

# Terminal domain example
messages = [
    {
        "role": "system",
        "content": "You are a language world
                    "Given the user's command

    },
    {
        "role": "user",
        "content": "Action: execute_bash\nCor
    }
]

response = client.chat.completions.create(
    model="Qwen/Qwen-AgentWorld-35B-A3B",
    messages=messages,

```

```
max_tokens=32768,  
temperature=0.6,  
)  
print(response.choices[0].message.content)
```

We provide **domain-specific world model system prompt templates** in [prompts/](#) of the GitHub repository for all 7 domains. These serve as general-purpose system prompts when using Qwen-AgentWorld as an environment simulator. Each domain folder contains a `system_prompt.txt` (world model system prompt) and a `judge_system_prompt.txt` (evaluation prompt).

Evaluate on AgentWorldBench

AgentWorldBench evaluates language world models by scoring each predicted environment observation on 5 dimensions: **Format**, **Factuality**, **Consistency**, **Realism**, and **Quality**.

Setup

```
# Clone the evaluation repository  
git clone https://github.com/QwenLM/Qwen-AgentWorld  
cd Qwen-AgentWorld  
  
# Download the benchmark  
huggingface-cli download Qwen/AgentWorldBench  
  
# Install dependencies  
pip install openai
```

Run Evaluation

The evaluation follows a three-step pipeline:


```
cd eval
```

```
# Step 1: Run world model inference
```

```
python eval.py infer \  
  --data-dir ../AgentWorldBench \  
  --model-base-url http://localhost:8000/v\  
  --model-name Qwen/Qwen-AgentWorld-35B-A3I\  
  --output-dir ./results
```

```
# Step 2: Run LLM judge scoring
```

```
export OPENAI_API_KEY="your-api-key"  
python eval.py judge \  
  --predictions ./results/predictions.json\  
  --judge-base-url https://api.openai.com/  
  --judge-model gpt-5.2-2025-12-11 \  
  --output-dir ./results
```

```
# Step 3: Aggregate and display scores
```

```
python eval.py score --predictions ./results,
```

Best Practices

- 1. Sampling Parameters:** We recommend temperature=0.6, top_p=0.95, top_k=20 for world model inference. The model uses thinking mode by default (<think>... </think>) to reason about environment state transitions before producing the predicted observation.
- 2. Adequate Output Length:** We recommend an output length of 32,768 tokens for most queries. For long, multi-step trajectories, you may increase the max output length to accommodate detailed environment observations.

3. **Domain-Specific System Prompts:** For optimal simulation fidelity, use the domain-specific system prompts provided in the [prompts/](#) directory of the GitHub repository.

Citation

If you find our work helpful, feel free to give us a cite.

```
@article{zuo2026qwen,  
  title={Qwen-agentworld: language world model  
    Yuxin and Xiao, Zikai and Shen  
    TOS Privacy About Careers  
    journal=arXiv preprint arXiv:2606.24597},  
  year={2026}  
}
```